ADITYA PATIL

Bengaluru, India

SUMMARY

High-achieving mathematics and STEM student with independent research experience in number theory, Diophantine equations, cryptography, and Hurwitz quaternions. Demonstrated impact through peer-reviewed publication, national and international research presentations, competitive mathematics distinctions, and leadership expanding access to advanced STEM learning.

EDUCATION

Class 12: CBSE, Global Indian International School, Whitefield, Bengaluru May '27

Grade 11: 92/100 | Global Citizen Scholar (full-ride scholarship)

Class 10: CBSE, Podar International School, Chalisgaon, Maharashtra May '25

Grade 10: 96/100 | Grade 9: 95/100 | CBSE National Topper in English: 100/100

TECHNICAL SKILLS

Mathematics & Research: Number theory, Diophantine equations, Galois theory, cryptography, Hurwitz quaternions, mathematical proof writing

Scientific & Product Development: AI-assisted prototyping, renewable-energy systems, field data collection, community testing

Financial Modeling: Private-equity valuation, three-statement modeling, leveraged buyout modeling, waterfall distributions

EXPERIENCE

University of Toronto (Online): Number Theory Research Jun '25 – Jan '26

Mentor: Prof. Ila Varma, Department of Mathematics

DOI:
//rsisinternational.org/journals/ijrsi/view/the-non-existence-of-integer-solutions-to-the-quartic-cubic-diophantine-equation-y-3-xy-x-4-4-a-complete-resolution-via-factorization-and-modular-arithmetic

- Proved the non-existence of integer solutions for an unsolved cubic-quartic Diophantine equation and examined related infinite families.
- Developed signing and verification algorithms demonstrating 99% security within the project evaluation framework, and proposed a framework for a new post-quantum cryptosystem.
- Published the research in the International Journal of Research and Science Innovation and presented it at the IRIS National Fair, earning a Silver Medal.
- Presented findings at the International Mathematics and Statistics Student Research Symposium and was selected to present at the International Conference of Number Theory and Diophantine Analysis in Mauritius.

Indian Statistical Institute: Hurwitz Quaternion Research Oct '25 – Present

Mentor: Prof. Ramesh Shreekantan, Department of Mathematics

- Investigate which Hurwitz integers can be represented as the sum of two Hurwitz primes, where a Hurwitz prime is an element of the maximal order with rational-prime reduced norm.
- Introduced a Norm Interval Theorem and formulated conjectures connected to ternary Goldbach-type questions.
- Presented the work at the International Mathematics and Statistics Student Research Symposium.
- Qualified among the Top 46 projects in the Asia Region during Round 1 of the S.-T. Yau High School Science Award, Mathematics category.

PROMYS India, IISc Bengaluru May '25 – Jun '25

- Completed rigorous coursework in number theory and Galois theory through proof-intensive problem sets, lectures, and collaborative exploration.
- Received the Mehta Fellowship for fully funded participation in the highly selective program.
- Returned as an alumnus to deliver a lecture titled “Additive Properties of Hurwitz Quaternions.”

Yale University: Yale Young Global Scholars (IST) Jul '26

Two-week on-campus program | Full-ride scholar

- Studied mathematics and medical sciences through seminars and interdisciplinary classes in the Innovation, Science & Technology track.
- Completed a capstone project examining the impact of artificial intelligence on cognitive ability.

ACADEMIC PROJECTS

SOLAR-SMART: Intelligent Solar-Energy Retrofit Device Aug '23 – Present

Team Lead | New York Junior Academy of Sciences Energy Infrastructure Challenge

Project Portfolio - https://drive.google.com/file/d/1o4sifzjBZ28RP_fcQ1tIHJ166_ywMHGT/view?usp=sharing

- Led a six-student team in developing a compact, low-cost retrofit device that uses the SER-1 engine to route energy automatically, protect battery health through smart thresholds, and alert users to performance drops.
- Designed the system to address dust accumulation and battery degradation, with the goal of recovering 20–30% more usable solar energy.
- Supported renewable-energy adoption for 30+ households through awareness initiatives, installation assistance, and guidance on subsidy access.
- Earned the Projects of Distinction award, placing in the Top 11 of 400 competing teams.

BreathNext: Portable Passive Bio-Panel for Indoor Air Quality Oct '25 – Present

Team Lead | Four-member school research team

Project - https://drive.google.com/drive/folders/1uSoANOW0_xid1Qqz0Nf0ba_zKPA2rYy-

- Led development of an AI-integrated synthetic-biology device designed to detect indoor pollutants and reduce wall dampness and respiratory-health risks.
- Conduct data collection and field research in collaboration with ASHA community members.
- Won the GIIS Young Scientist Challenge for the project's affordable respiratory-health platform.
- Qualified for Round 1 of the S.-T. Yau High School Science Award, Asia Region, Biology category.

OTHER WORK EXPERIENCE

Motilal Oswal: Investment Analyst Intern Apr '24 – May '24

Six-week residential internship | Only high-school intern in the cohort

- Built a mathematical valuation model for private-equity investments that the team adopted for live deal screening.
- Developed three-statement, leveraged buyout, and waterfall-distribution models to evaluate investment outcomes and capital structures. Earned a letter of recommendation for performance and contribution during the internship.

LEADERSHIP & INITIATIVES

Xponentia: Founder Dec '24 – Present

- Founded an initiative providing high-school students, particularly those from under-resourced communities, access to research guidance, mentorship, and foundational learning resources.
- Reached 1,000+ students across 20+ schools through educational outreach supporting independent scientific inquiry.
- Manage a 7,000-member Telegram community that evolved from Olympiad support into a broader mathematics and STEM learning network.

SEVA under Pledge a Smile Foundation Dec '23 – Present

Project Lead, Fundraiser & Social Media Manager

- Advanced from core volunteer to Social Media and Fundraising Intern based on sustained contribution.
- Raised USD 2,500 through targeted digital-marketing, social-media, and fundraising campaigns.
- Organized three on-the-ground food drives and created posters, stickers, reels, advertisements, and campaign materials for vulnerable communities.

Alpha Pi-rates Mathematics Club Jan '23 – Present

- Founded the school's first mathematics club and built a 22-member community of students and teachers.
- Guide peers in proof techniques, advanced problem-solving, and preparation for mathematics competitions.
- Represented the school in mathematics meets including the Stanford Math Tournament and Purple Comet! Math Meet.

AWARDS & RECOGNITIONS

Major Distinctions

- Silver Medal, IRIS National Fair — Ranked 2nd among 15,000 submissions (Top 0.013%) in the Mathematics category for independent research in pure mathematics and cryptography.
- S.-T. Yau High School Science Award — Qualified Round 1 in Mathematics and Biology; Mathematics project ranked among Asia's Top 46.
- Indian National Mathematical Olympiad Qualifier — Top 900 of 100,000+ competitors nationwide after clearing IOQM and RMO.

- Global Citizen Scholarship, GIIS — One of 10 students selected from 11,000 applicants for a full-ride scholarship..
- First Place, India, Purple Comet! Math Meet — Scored 30/30 for first place in India in the Small High School category.
- Stanford Math Tournament — Received Honorable Mention twice, in Number Theory and Calculus.
- Dual Presenter, IMSSRS Symposium — Selected for live and asynchronous presentations in Diophantine equations, cryptography, and quaternions.

Additional Honors

- All India Rank 16, TALLENTEX 2024 — Scored 300/320 and earned a 90% scholarship in the national STEM talent search.
- All India Rank 24, PW-NSAT — Ranked in the Top 0.01% nationally, receiving a 90% scholarship and cash award.
- 100% Tuition Scholarship, FIITJEE Talent Reward Exam — Scored 220/275 and earned a complete tuition waiver.
- Jawahar Navodaya Vidyalaya Selection Test — Selected for a fully funded seven-year government scholarship.
- International Rank 12, SOF Mathematics Olympiad — Achieved International Rank 12 in 2024 and Rank 150 in 2023.
- National Finalist, SilverZone Olympiads — Level 3 qualifier in Mathematics and English; Level 2 in Science.
- Top 1% Nationally, TERI Green Olympiad — Two-time Certificate of Excellence recipient in 2023 and 2024.
- CBSE & Institutional Honors — Scored 100/100 in Class 10 English; received Academic Excellence, Math Maestro, Best English Communicator, and GIIS Young Scientist awards.